GCSE Maths Paper 3 (Calculator) · Wed 10 June · 9:00 AM

The Paper 3 Calculator Hit-List

Both tiers · AQA · Edexcel · OCR · WJEC

Paper 3 is the **last** maths paper, so every topic is fair game — which is exactly why people panic-revise the wrong things. This is the short list: the calculator-paper topics that come up almost every year and decide the most marks. Foundation must-haves first, then the top-of-Higher band that separates a 7 from a 9.

AQA Paper 3 | Edexcel Paper 3 | OCR Paper 3 / Paper 6 | WJEC Unit 2

What to revise in the last days before Wednesday

Calculator-heavy, high-frequency, and easy to lose marks on if you're rusty. Don't try to relearn everything — lock these down.

FOUNDATION — LOCK THESE IN

1 **Percentages: increase, decrease & reverse** *use the calculator*

Multipliers do it in one step: +15% is $\times 1.15$, −20% is $\times 0.8$. **Reverse percentage** is the trap — "a price is £48 after a 20% off, find the original" means $\div 0.8$, never ×1.2.

2 **Compound interest & growth/decay** *use the calculator*

Use the power: amount $= \mathrm{P} \times (\mathrm{multiplier})^{\mathrm{n}}$. The whole point of a calculator paper. Watch for "how many years until it exceeds…" — test values of n.

3 **Ratio: sharing, scaling & "given one part"**

Find the value of one share first. The hard version gives you the **difference** between two shares ("A gets £12 more than B") — that difference is the number of parts difference.

4 **Compound measures: speed, density, pressure**

Speed = distance ÷ time, density = mass ÷ volume, pressure = force ÷ area. **Unit conversion** is where marks vanish — convert to consistent units (e.g. km/h ↔ m/s) before you divide.

5 **Pythagoras & right-angled trig (SOH-CAH-TOA)** *degrees mode*

Find a missing side or angle in a right-angled triangle. To find an **angle**, use $\sin^{-1}$ / $\cos^{-1}$ / $\tan^{-1}$. Check the calculator shows a small **D** for degrees.

6 **Probability: tree diagrams & relative frequency**

Multiply **along** branches, add **between** outcomes. “At least one” = 1 − P(none). Expected number = probability × number of trials.

HIGHER — WHERE 7S BECOME 9S

1 **Sine rule, cosine rule & area = ½ab·sinC** on the formula sheet

Non-right-angled triangles. Cosine rule for a side from two sides + the included angle; rearranged cosine rule for an angle from three sides; ½ab·sinC for area. Often hidden inside a bearings or 3D question.

2 **Vectors: collinearity & ratio proofs**

The one most people drop. Express a route in terms of **a** and **b**, then prove two vectors are parallel (one is a scalar multiple of the other) or that points are collinear. **Full worked example below.**

3 **Upper & lower bounds in context**

Don’t just write the error interval — push it to a conclusion. Find the upper and lower bound of a calculation, then state the answer to a suitable degree of accuracy (the digits that agree on both bounds). Quotients: max = max numerator ÷ **min** denominator.

4 **Iteration** use ANS

Two skills. First, show a root lies between two values using a **sign change**. Then run $x_{n+1} = f(x_n)$ with the **ANS** key until consecutive iterates agree to the required d.p.

5 **Direct & inverse proportion (algebraic)**

$y \propto x^3$, $y \propto 1/\sqrt{x}$ and similar. Find k from the given pair, write the full equation, then solve. The square / cube / root forms are the Higher trap.

6 **Rates of change & area under a graph**

Speed–time graphs: area under = distance; gradient at a point = acceleration. Estimate area with the trapezium rule; estimate gradient by drawing a tangent. State units, and say if it’s an over- or under-estimate.

7 **Histograms & frequency density**

Frequency = frequency density × class width. Find the frequency in part of an unequal-width bar; estimate the median or a proportion above a given value.

8 **3D Pythagoras & trig + volumes of cones/spheres** on the formula sheet

Find the longest diagonal of a cuboid or the angle between a line and a plane. Volume/surface area of spheres, cones and frustums — the formulas are **given**, so the marks are in setting it up, not recalling it.

⭐ Worked exemplar: Vectors proof (the question most people drop)

> *OAB is a triangle with* $OA = a$ *and* $OB = b$*. P is the point on AB such that* $AP : PB = 1 : 2$*. Q is the point such that* $BQ = 2a$*. Prove that O, P and Q lie on a straight line. Work the engine, not the wording.*

Step 1. Write every vector in terms of a and b only.

Travel along known arrows; reversing an arrow makes it negative.

- $\mathbf{AB} = OB - OA = b - a$
- $\mathbf{AP} = ⅓·AB = ⅓(b - a)$ (P is ⅓ of the way along AB)
- $\mathbf{OP} = OA + AP = a + ⅓(b - a) = ⅓(2a + b)$
- $\mathbf{OQ} = OB + BQ = b + 2a = 2a + b$

Step 2. The step everyone forgets: factor out a scalar and compare.

- $\mathbf{OP} = ⅓(2a + b)$ and $\mathbf{OQ} = 2a + b$
- So $\mathbf{OQ} = 3 \times ⅓(2a + b) = 3·\mathbf{OP}$, so OQ is a scalar multiple of OP.

Step 3. State the conclusion in words.

- $\mathbf{OQ} = 3·\mathbf{OP}$, and the two vectors share the point O.
- Therefore O, P and Q lie on a straight line, with $OP : PQ = 1 : 2$.

CONCLUSION MARK The final mark is almost always for the **words**: “**OQ** is a scalar multiple of **OP**, they share the point O, therefore O, P, Q lie on a straight line.” Do the algebra without that sentence and you lose the conclusion mark. The scalar (here **3**) gives you the **ratio**.

Three calculator habits that save marks on Paper 3

- **Type compound expressions in one go** using brackets. Never round a middle step then re-enter it. Rounding early is the #1 silent mark-killer on the calculator paper.
- **Check the mode** before any trig: a small **D** at the top of the screen means degrees.
- **Keep π and surds exact** until the final line, then round only the answer to the accuracy the question asks for.

Free to share with anyone sitting Wednesday. Good luck — you’ve got this.